QuercusHealth AI: Automated Detection and Health Classification of *Quercus ilex* in the Spanish Dehesa via Domain-Adapted Aerial Imagery

Sergio Illescas Cabiro
Illinois Institute of Technology
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sillescascabiro@hawk.illinoistech.edu

Abstract

The Spanish Dehesa, a UNESCO-protected mosaic of holm oaks spanning roughly five million hectares across the Iberian Peninsula, is under sustained attack from *La Seca*, a root disease caused by the water mold *Phytophthora cinnamomi* [1]. Manual ground-level monitoring at this scale is impractical. We present QuercusHealth AI, a complete deep-learning pipeline for automated detection and health classification of *Quercus ilex* crowns from high-resolution nadir satellite imagery. Our approach adapts DeepForest [10], a RetinaNet-based crown detector pre-trained on millions of North American canopy images, to the visually distinct Spanish Dehesa through domain-specific fine-tuning. We first quantify the domain shift statistically (Kolmogorov-Smirnov $D = 0.86$, $p < 0.0001$), then address it through a five-phase pipeline. A two-stage architecture combining a fine-tuned one-class tree detector with a ResNet-18 crop classifier achieves a macro-averaged F1 of 0.636 end-to-end and a Seca-class F1 of 0.346, compared to 0.000 under the zero-shot baseline. These results demonstrate that careful transfer-learning methodology can support scalable ecological monitoring over geographies too vast for human surveyors. Source code, trained model weights, and the annotated dataset are publicly available.[1]

1. Introduction

The Dehesa is one of Europe's most biodiverse land-cover types, a centuries-old silvopastoral system of open woodland and pasture dominated by holm oak (*Quercus ilex*) in southwestern Spain and Portugal. Beyond its ecological significance, it functions as a critical carbon sink, a reservoir for native fauna, and the productive base of the Iberian acorn-fed pig industry. Since the 1980s, the ecosystem has faced an accelerating dieback known colloquially as *La Seca*. The primary pathogen is *Phytophthora cinnamomi*, a water-borne oomycete that colonizes root systems, blocks water uptake, and causes progressive crown thinning followed by death [3]. By the time a land manager observes visible symptoms from the ground, the infected root system is already beyond recovery.

Satellite and airborne remote sensing offer a path toward early detection at landscape scale. Modern nadir imagery at sub-meter resolution can resolve individual tree crowns clearly enough to distinguish the dense, circular canopy of a healthy holm oak from the sparse, radiating silhouette characteristic of *La Seca*. However, directly applying a generic object detector trained on different forest types introduces substantial domain shift, because pixel statistics, soil background, crown density, and illumination differ sharply between Dehesa and the forests on which most pre-trained models were developed.

In this work we build and evaluate a five-phase pipeline: systematic data collection via automated satellite image capture, multi-temporal ground-truth annotation, zero-shot baseline evaluation with statistical domain-shift analysis, multi-class fine-tuning, and a two-stage detection-plus-classification architecture. Our main contributions are (1) a replicable data acquisition strategy for remote Dehesa monitoring, (2) a statistically grounded proof of domain shift for this ecosystem, and (3) a two-stage deep-learning system that achieves non-trivial Seca-class detection where a direct fine-tuned detector yields zero recall.

2. Related Work

Tree crown detection. Automatic delineation of individual tree crowns from aerial imagery has been studied for several decades using rule-based segmentation and later deep-learning detectors [5]. The most directly relevant prior work is DeepForest [10], which releases a RetinaNet model pre-trained on over 10^6 annotated tree crowns from the NEON airborne observatory [9, 11]. Because NEON covers temperate North American forests, its feature space captures trunk-to-canopy ratios and crown shapes that transfer well to many closed-canopy scenarios. The sparse, bright-soil Dehesa, however, sits far outside that distribution, motivating the domain-adaptation experiments described in Section 4.

Object detection architectures. RetinaNet [7] addresses the extreme foreground/background imbalance inherent in dense object detection through Focal Loss, which down-weights easy negatives and concentrates gradient energy on hard examples. The backbone is a ResNet [4] with a Feature Pyramid Network [6] that aggregates multi-scale features.

[1]Code: https://github.com/sergioillescascabiro/QuercusHealth-Public Model: https://huggingface.co/sillescas/deepforest-dehesa-quercus Dataset: https://app.roboflow.com/ds/lhbhwAM9HN?key=xqoQlM4K1Y

For our secondary classification stage we use ResNet-18, pre-trained on ImageNet [2], which provides strong low-level texture and color features relevant to distinguishing healthy from diseased foliage.

Transfer learning and catastrophic forgetting. A well-known risk in fine-tuning pre-trained networks is catastrophic forgetting [8]: a learning rate that is too large causes gradient updates to overwrite the weights encoding previously acquired representations. We confirm this empirically in an ablation study (Section 5.3) and motivate our choice of a conservative learning rate of 10^{-4}.

3. Dataset and Data Acquisition

Image capture. All imagery was collected from Google Earth Pro using an automated capture script built on PyAuto-GUI. The script traverses an 18×18 grid in a boustrophedon (zig-zag) pattern over a single Dehesa estate in Extremadura, Spain, capturing one 800×800 pixel screenshot per grid cell at zoom level 19, which corresponds to approximately 0.5 m per pixel. Three seasonal capture sessions were conducted: summer 2019, winter 2021, and winter 2024, yielding 324 tiles per session (972 tiles in total).

Multi-temporal ground-truth strategy. A key challenge in labeling diseased trees from a single image is distinguishing *La Seca* crown loss from natural seasonal defoliation. We resolved this ambiguity by comparing the same geographic coordinate across the summer 2019 and winter 2024 images. A crown that appears sparse in 2019 and has vanished entirely or collapsed by 2024 is confirmed as a Seca casualty; a crown that merely thins in winter and recovers in a later summer image is labeled Healthy. Figure 1 illustrates this comparison: the annotated boxes in the 2019 tile mark crowns exhibiting early decline, and the same location in 2024 reveals bare ground where those trees once stood. This multi-temporal validation produces ground-truth labels verifiable without field visits.

Annotation and dataset statistics. Bounding-box annotation was performed for the summer 2019 capture session. A pre-annotation step (Phase 2b) ran the zero-shot DeepForest model at a low confidence threshold to generate candidate boxes, which were then reviewed and corrected by a human annotator: false positives removed, missed crowns added, and each crown labeled as *Healthy* or *Seca*. The finalized dataset contains 278 images split into 243 training, 18 validation, and 17 test images. The training partition holds 7,341 annotated bounding boxes (6,449 Healthy and 892 Seca), yielding a class ratio of 87.8% Healthy to 12.2% Seca. Validation and test sets contain 527 and 513 boxes respectively. Class imbalance is addressed at the architecture level via Focal Loss in the RetinaNet detector and via a weighted sampler together with class-weighted cross-entropy in the ResNet-18 classifier.

4. Methodology

Our pipeline progresses through five phases, each building on the results of the previous one.

4.1 Phase 1 and 2: Zero-Shot Baseline and Evaluation Framework

We began by evaluating DeepForest [10] without any fine-tuning on both NEON benchmark imagery and Dehesa test tiles. On NEON, the model achieves F1 $= 0.68$, confirming its strong in-distribution performance. On Dehesa, precision drops to 0.512 and recall to 0.230, yielding F1 $= 0.320$. To verify that this degradation constitutes genuine domain shift rather than random noise, we applied two non-parametric statistical tests to the prediction confidence score distributions. The two-sample Kolmogorov-Smirnov test yields $D = 0.86$, $p < 0.0001$, and the Mann-Whitney U test confirms the distributions are stochastically distinct ($p < 0.0001$). Figure 2 illustrates the two score distributions. These results provide the quantitative justification for domain adaptation.

A hyperparameter sweep over confidence thresholds (score_thresh $\in \{0.05, 0.10, 0.15, 0.20\}$) showed that threshold tuning alone can recover at most F1 $= 0.527$, well below the NEON baseline of 0.68, confirming that inference-time adjustments are insufficient and that model-level adaptation is required.

4.2 Phase 3: Fine-Tuning the Detection Backbone

We fine-tune the full DeepForest model end-to-end for two-class prediction (Healthy / Seca). The base model is loaded from the `weecology/deepforest-tree` HuggingFace checkpoint, which encodes RetinaNet [7] with a ResNet-50 [4] backbone and a Feature Pyramid Network [6] (32.1 million parameters). Fine-tuning replaces the single-class detection head with a two-class equivalent and trains the entire network.

Learning rate selection. We use LR $= 10^{-4}$, ten times lower than the DeepForest default. This choice is motivated by the risk of catastrophic forgetting [8]: the NEON pre-trained backbone encodes a strong inductive bias for circular, elliptical crown textures that is essential for detecting trees in any domain. At LR $= 10^{-2}$, the 315 gradient steps executed over 5 ablation epochs shift the backbone weights far enough from their pre-trained values that the model loses crown-detection capability entirely (ablation F1 $= 0.000$; see Section 5.3). At LR $= 10^{-4}$, the 1,890 steps over 30 epochs shift the feature space toward Dehesa-specific appearance (brighter sandy soil, sparser canopy) while preserving crown geometry.

Training configuration. Training ran on an NVIDIA RTX 3060 (12 GB VRAM) on a dedicated Ubuntu server, completing 30 epochs with batch size 4 in approximately 20 minutes. The IoU threshold for matching predictions to ground-truth boxes is set to 0.4, consistent with the DeepForest default for satellite imagery where bounding-box annotation precision is limited by pixel resolution. Dynamic data aug-

(a) Summer 2019: annotated crowns showing early signs of decline (sparse canopy, radiating branches). Purple bounding boxes mark trees under observation.

(b) Winter 2024: the same geographic area five years later. Trees that were annotated as suspect in 2019 have vanished or collapsed, confirming *La Seca* as the cause of death.

Figure 1. Multi-temporal ground-truth validation. Comparing the same coordinates across years allows reliable labeling of Seca trees without physical field visits.

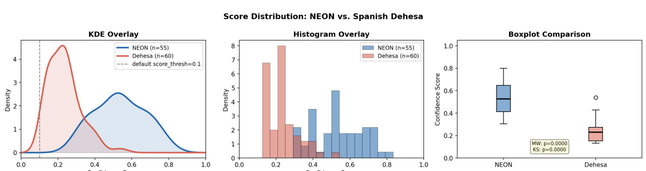

Figure 2. Confidence score distributions from the zero-shot DeepForest model on NEON benchmark imagery versus Dehesa tiles. Mean confidence drops from 0.535 on NEON to 0.233 on Dehesa, a 56% reduction. The KS test ($D = 0.86$, $p < 0.0001$) provides statistical evidence of domain shift.

mentation was applied per epoch via the `albumentations` library: horizontal flip ($p = 0.5$), vertical flip ($p = 0.5$), random brightness and contrast perturbation ($p = 0.3$, range ± 0.2), hue-saturation-value shift ($p = 0.2$), and Gaussian noise ($p = 0.1$). Focal Loss (built into RetinaNet) handles class imbalance natively.

4.3 Phase 4: Two-Stage Classification Pipeline

Although Phase 3 fine-tuning raises the overall F1 substantially, it fails to produce non-zero Seca recall because the joint optimization of detection and two-class discrimination overwhelms the gradient signal for the minority Seca class. We therefore decouple detection from classification in a two-stage pipeline.

Stage 1 uses the DeepForest zero-shot detector (single class: Tree) to propose candidate crown regions. Stage 2 extracts 96×96 pixel crops centered on each proposed box (with 8-pixel padding) and passes them to a ResNet-18 binary classifier [4, 2]. The classifier replaces the standard global average pooling head with a $512 \rightarrow 256 \rightarrow$ReLU$\rightarrow$Dropout(0.4)$\rightarrow 2$ linear chain. Training uses a WeightedRandomSampler to balance class exposure per batch and a class-weighted cross-entropy loss. The first five epochs freeze the backbone while only the custom head is trained (warmup phase), after which the full network is unfrozen at one-tenth the base learning rate. Dynamic augmentation via `torchvision` includes random horizontal and vertical flips, random rotation up to 20°, and color jitter on brightness, contrast, saturation, and hue. Training ran for 30 epochs with batch size 64.

4.4 Phase 5: End-to-End One-Class Pipeline

The two-stage Phase 4 system was evaluated with ground-truth crops, giving an optimistic upper bound on classifier performance. Phase 5 closes this gap by replacing the zero-shot Stage 1 detector with a DeepForest model fine-tuned on a single collapsed class (*Tree*), trained for 20 epochs with the same configuration as Phase 3. The one-class objective simplifies the detection task and improves localization quality. The ResNet-18 classifier from Phase 4 is reused without retraining, completing the fully automated end-to-end pipeline.

5. Experiments and Results

5.1 Domain Shift Analysis

The statistical analysis in Phase 1 establishes a quantitative baseline for the domain discrepancy. Confidence scores from the zero-shot model on NEON imagery follow a unimodal distribution centered near 0.535, whereas Dehesa scores cluster

Table 1. Test-set performance across pipeline phases. Phase 4 evaluates the classifier on ground-truth crops; Phase 5 evaluates the full end-to-end pipeline.

Phase	F1 (All)	F1 (Seca)	Prec / Rec
2: Zero-shot	0.320	0.000	0.51 / 0.23
3: Fine-tuned	0.669	0.000	0.61 / 0.74
4: Two-stage (GT)	0.712	0.354	n/a
5: End-to-end	0.636	0.346	n/a

below 0.25 (Figure 2). The KS-statistic of 0.86 indicates that 86% of the score range separates the two cumulative distribution functions, a magnitude of shift that rules out sampling coincidence. This analysis provides the scientific justification for every fine-tuning decision that follows.

5.2 Quantitative Results

Table 1 summarizes performance across all pipeline phases, evaluated on the held-out test partition of 17 images and 513 annotated boxes. Phases 2 and 3 report results from the end-to-end detection model; Phases 4 and 5 report results from the full two-stage system.

Fine-tuning in Phase 3 raises overall F1 from 0.320 to 0.669, a gain of 34.9 percentage points, while Seca recall remains zero. The two-stage design in Phase 4 achieves Seca F1 of 0.354 on ground-truth crops, breaking the Seca detection barrier for the first time. Phase 5 incurs a localization tax of 0.8 percentage points on Seca F1 (0.354 to 0.346) when Stage 1 runs fully automatically, confirming that the ResNet-18 classifier is robust and that the primary source of end-to-end error is localization, not classification.

5.3 Ablation Study

To isolate the effect of learning rate on catastrophic forgetting, we ran a controlled ablation with $\text{LR} = 10^{-2}$ for 5 epochs using the same data and architecture as Phase 3. The result is a complete collapse to $\text{F1} = 0.000$: the model predicts no boxes above the confidence threshold. Figure 3 visualizes this contrast: the high learning rate produces zero useful predictions while $\text{LR} = 10^{-4}$ yields $\text{F1} = 0.669$.

The training dynamics at $\text{LR} = 10^{-4}$ are shown in Figure 4: both training and validation loss decrease monotonically over 30 epochs with a small but stable generalization gap, indicating that the model has not overfit. Loss curves for the high-LR ablation show steeper initial descent followed by divergence, confirming the theoretical prediction that fine-tuning a large pre-trained network on a small domain-specific dataset requires a conservative learning rate.

5.4 Qualitative Analysis and Failure Cases

Figure 5 presents a side-by-side qualitative comparison on a representative test tile. The baseline model misses most crowns in the Dehesa domain and generates frequent false

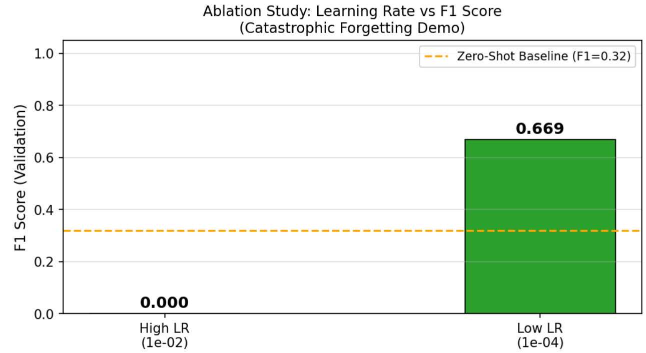

Figure 3. Ablation study: learning rate 10^{-2} causes catastrophic forgetting (F1 = 0.000); learning rate 10^{-4} achieves F1 = 0.669.

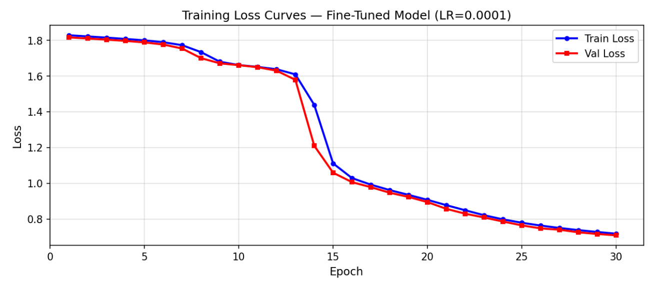

Figure 4. Training and validation loss over 30 epochs at LR = 10^{-4}. Both curves decrease monotonically; the small gap indicates moderate regularization without overfitting.

positives on the bright ochre-colored soil. After fine-tuning, crown localization improves markedly.

The confusion matrices reveal a more nuanced picture. The Phase 3 detector (Figure 6) correctly classifies Healthy crowns with high precision but fails entirely on Seca. The Phase 4 ResNet-18 classifier (Figure 7) successfully redistributes attention to Seca, achieving non-zero recall at some cost to Healthy precision.

The primary failure modes are threefold. First, the NMS threshold (nms_thresh $= 0.05$) is aggressive: when holm oak crowns overlap or cluster at the margins of the estate, NMS suppresses legitimate secondary detections, producing false negatives in dense areas. Second, very small shrubs and large stones share spectral characteristics with sparse Seca crowns, generating persistent false positives. Third, tiles captured at low solar elevation angles produce elongated shadows that partially occlude crown boundaries, reducing IoU with ground-truth boxes below the 0.4 acceptance threshold.

Figure 8 shows the complete Phase 5 end-to-end output on a raw test tile, where Stage 1 localizes tree positions and Stage 2 assigns health labels to each detected crop.

6. Conclusion

We have presented QuercusHealth AI, a five-phase pipeline that adapts a North American forest canopy detector to the

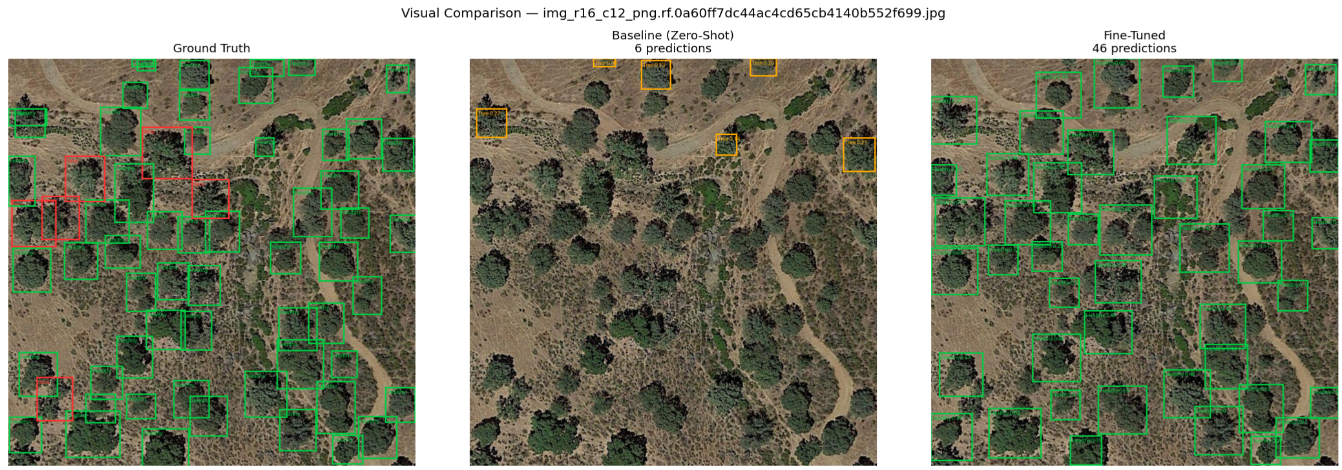

Figure 5. Qualitative comparison on a representative test tile. *Left*: ground-truth annotations (green for Healthy, red for Seca). *Center*: zero-shot DeepForest baseline predictions. *Right*: Phase 3 fine-tuned predictions. The baseline misses most crowns and produces scattered false positives on the bright soil. After fine-tuning, the detector correctly localizes the majority of crowns.

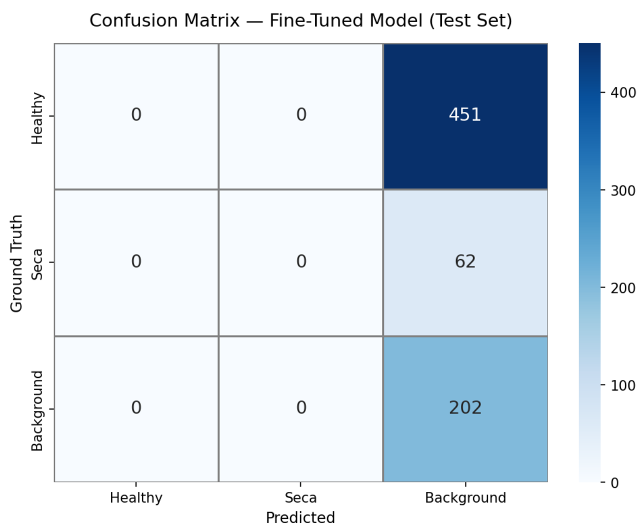

Figure 6. Phase 3 fine-tuned detector confusion matrix. Seca recall is zero: the joint detection-classification objective overwhelms the gradient signal for the minority class.

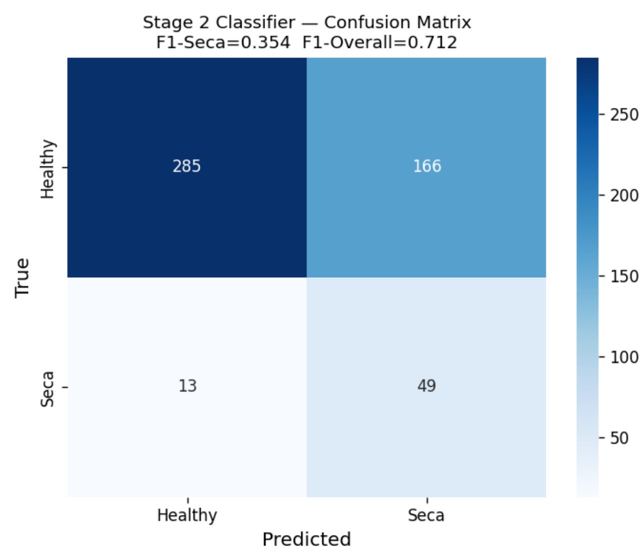

Figure 7. Phase 4 ResNet-18 classifier confusion matrix on ground-truth crops. Seca recall reaches 0.354, confirming that decoupling detection from classification is essential for minority-class detection.

Spanish Dehesa for the purpose of automated *Quercus ilex* health classification. Our work makes three technical contributions: a systematic proof of domain shift using non-parametric statistical tests, a fine-tuning protocol that avoids catastrophic forgetting through conservative learning rate selection, and a two-stage architecture that achieves non-zero Seca detection (F1 = 0.346 end-to-end) where a direct fine-tuned detector cannot. The pipeline processes one 800×800 pixel tile in under one second on commodity hardware, suggesting that a scheduled sweep of Extremadura's 5 million hectares of Dehesa is operationally feasible with available compute.

Future work should focus on three directions: (1) expanding the annotated dataset beyond the current 278 images to improve Seca class coverage and reduce over-reliance on the minority-class weighting strategies; (2) exploring instance segmentation methods such as Mask R-CNN to produce crown polygons instead of bounding boxes, which would enable crown-area change tracking across seasons; and (3) replacing static Google Earth captures with scheduled API calls to satellite data providers, enabling fully automated multi-temporal monitoring without manual screen capture.

Author Contributions

Sergio Illescas Cabiro conceived the project, designed the automated satellite capture pipeline, performed all annotations,

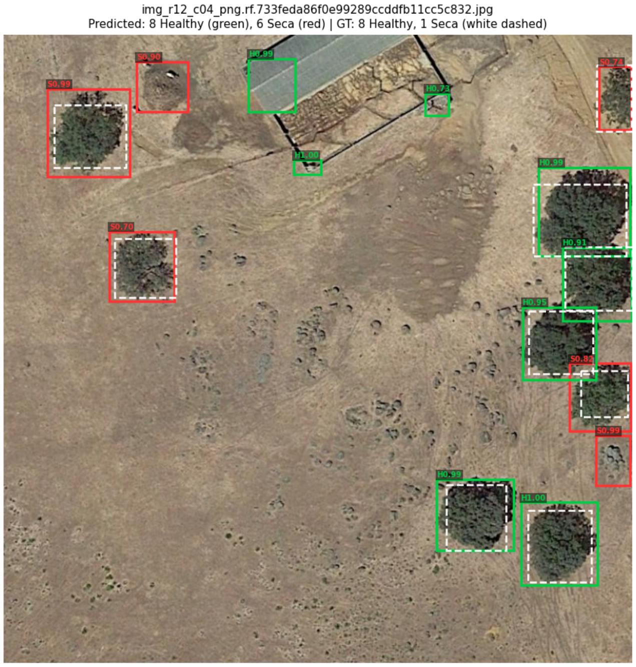

Figure 8. End-to-end Phase 5 output on a raw test tile. Green boxes indicate predicted Healthy crowns; red boxes indicate predicted Seca. Stage 1 localizes tree positions; Stage 2 assigns health labels.

implemented all five pipeline phases, conducted training on the GPU server, analyzed results, and wrote this report.

AI Use Statement

Claude (Anthropic) was used for assistance with code debugging, LaTeX formatting, and prose revision. All experimental design, training decisions, result interpretation, and scientific conclusions are the sole work of the author.

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